Trends in Biotechnology

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Review

Engineered macromolecular Toll-like receptor agents and assemblies

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Macromolecular Toll-like receptor (TLR) agents have been utilized as agonists and inhibitors in preclinical and clinical settings. These agents interface with the TLR class of innate immune receptors which recognize macromolecular ligands that are characteristic of pathogenic material. As such, many agents that have been historically investigated are derived from the natural macromolecules which activate or inhibit TLRs. This review covers recent research and clinically available TLR agents that are macromolecular or polymeric. Synthetic materials that have been found to interface with TLRs are also discussed. Assemblies of these materials are investigated in the context of improving stability or efficacy of ligands. Attention is given to strategies which modify or enhance the current agents and to future outlooks on the development of these agents.

Highlights

Toll-like receptors (TLRs) serve as drug targets for a variety of indications ranging from cancer to autoimmune disease.

Biomaterials of both natural and synthetic origin have been increasingly investigated as TLR-interfacing agents for treating disease, some of which have been approved by the FDA as vaccine adjuvants.

Engineering strategies which modify or harness these compounds provide new avenues to enhance drug delivery, receptor activation, and downstream immune activity.

Strategies to modulate the innate immune system rely on a more nuanced approach, tailoring materials chemistry and biophysical pattern display to produce differential immune responses. These strategies often rely on a bottom-up understanding of TLR activity.

Macromolecular TLR agents

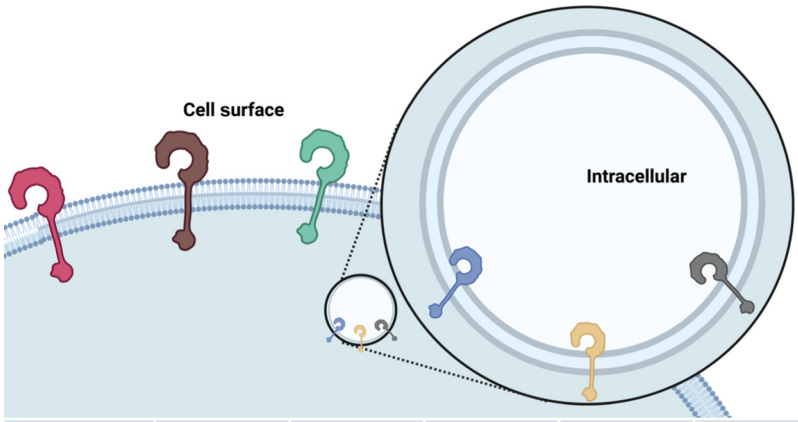
Historically, **biomaterials** (see Glossary) have been developed to minimize the host immune response. However, in recent years interest has arisen in engineering immunomodulatory biomaterials to tailor the immune response, especially those that target **TLRs**. TLRs serve as part of the body's first line of defense against bacteria, viruses, and fungi. TLRs are transmembrane proteins that are found primarily on **innate immunity**cells – that is, macrophages, neutrophils, dendritic cells (DCs) (Figure 1) – that have specialized recognition of multiple ligand classes, including nucleic acids, lipopeptides (LPs), and lipopolysaccharide (LPS). These ligands contain **pathogen-associated molecular patterns (PAMPs)**, which are chemical and structural motifs characteristic of pathogenic **macromolecules** (Figure 2). Upon activation, TLRs initiate signaling pathways which lead to the upregulation of proinflammatory cytokines. As gatekeepers of the human innate immune system (Figure 1), TLRs are an important therapeutic target for inflammatory disease, cancer immunotherapies, and subunit vaccines [1,2].

While there are numerous small-molecule TLR **agonists**, there is broad interest in macromolecular biomaterial strategies which target TLR activity [3–5]. Macromolecules can interface directly with TLRs to agonize or antagonize activity, or may indirectly influence their behavior via PAMP scavenging (Table 1, Key table; Figures 3 and 4). These include synthetic and biological polymers as well as other large hydrocarbon-based structures such as lipid assemblies (Figure 3). Their larger size enables the display of chemical and physical patterns which play an advantageous role in interfacing with pattern recognition receptors (PRRs) while minimizing side effects that are often associated with small-molecule drugs [6]. That is, macromolecules and assemblies can incorporate multiple copies of TLR agonists and antagonists, and control their spatial and temporal presentation. Synthetic and bioinspired macromolecules serve as an advantageous starting place for developing new TLR technologies, as they are easily reproduced, may avoid off-target recognition, and can be assembled into higher-order structures that maintain nanoscale control [7].

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	TLR2	TLR4	TLR5	TLR3	TLR7/8	TLR9
Cell type	• Dendritic • Monocyte • Macrophage • B cell • T cell • Microglial	•Dendritic •Monocyte •Macrophage •B cell •Neutrophil •Mast	•Monocyte •Macrophage •T cell •Keratinocyte	•Dendritic •Macrophage •T cell •Microglial •NK cell	•pDC, mDC •Monocyte •Macrophage •B cell	•pDC •Monocyte •Macrophage •B cell
Ligand	•Glycolipids and proteolipids of pathogenic cell wall	•LPS •Host polysacch-arides and glycoproteins	•Flagellin	•dsRNA	•ssRNA	•Unmethylated CpG DNA
Down stream effects	•Inflammatory cytokines (TNF-α, IL-1β, IL-6, IL-10, IL-12)	•Inflammatory cytokines (TNF-α, IL-1β, IL-6, IL-10, IL-12) •Type I IFN	•Inflammatory cytokines (KC, IL-6, G-CSF) •Chemotactic effects •T cell stimulation	•Inflammatory cytokines •Type I IFN (IFN-β and IFN-α) •IL-6 •T cell stimulation •Chemotactic effects	•Inflammatory cytokines •Type I IFN (TNF-αt, IL12p40)	•Inflammatory cytokines (IL-1β, IL-6, IL-10, IL-12) •Type I IFN

Trends in Biotechnology

Figure 1. Toll-like receptors (TLRs), expressed either on the cell surface or intracellularly, are activated by various ligands, resulting in increased expression of inflammatory cytokines. Cell surface TLRs include TLR2, TLR4, and TLR5. These are all expressed by a wide range of immune cells. While TLR2 is primarily responsible for detecting pathogen-associated molecular patterns (PAMPs), TLR4 recognizes lipopolysaccharide (LPS) and envelope proteins. Finally, TLR5 binds to bacterial flagellin, leading to secretion of cytokines and chemoattraction of immunocytes to help eliminate bacteria. Intracellular TLRs include TLR3, TLR7, TLR8, and TLR9. TLR3 is primarily expressed by dendritic cells, macrophages, NK cells, and T cells, where it recognizes double-stranded RNA (dsRNA) that is produced during replication of RNA viruses following infection. TLR7, TLR8, and TLR9 are expressed by plasmacytoid dendritic cell (pDCs), B lymphocytes, monocytes, and macrophages, and contribute to the immune system's ability to eliminate viral and bacterial pathogens. TLR7 and TLR8 both play key roles in the antiviral response, as they recognize viral single-stranded RNA (ssRNA) and imidazoquinolines. While both of these TLRs respond to the RNA ligands, their signaling pathways and downstream effects may depend on cell type and activating ligand. Finally, TLR9 binds to unmethylated DNA with CpG motifs, most commonly associated with bacterial DNA. Abbreviations: G-CSF, granulocyte colony stimulating factor; IFN, interferon; IL, interleukin; KC, keratinocyte-derived chemokine; mDC, myeloid dendritic cell; NK cell, natural killer cell; TNF-α, tumor necrosis factor α. Figure created with BioRender.

Glossary

Agonist: a compound which activates a receptor to produce a biological response (i.e., a proinflammatory immune response).
Biomaterial: any substance (other than drugs) or combination of substances that is natural or synthetic in origin which can be used for any period of time, as a whole or as a part of a system which treats, augments, or replaces tissues, organs, or functions of the body.
Glycosaminoglycan (GAG): a class of sulfated polysaccharides expressed in the extracellular matrix of human cells.
Innate immunity: the more ancient arm of the immune system, which initiates and orchestrates a quick proinflammatory response to pathogens in the body.
Macromolecule: a compound that is composed of many atoms, particularly carbon atoms.
Molecular weight (MW): the mass of a molecule or macromolecule, often expressed in g/mol. The MW of a polymer reflects its degree of polymerization.
Nuclear factor κB (NF-κB): a protein transcription factor that signals downstream from most TLRs. Expression of this factor is often linked to TLR activation.
Oligodeoxynucleotide (ODN): a short, synthetic DNA ligand, typically containing unmethylated CpG motifs that are activators of TLR9.
Pathogen-associated molecular patterns (PAMPs): chemical and structural motifs that are characteristic of pathogenic macromolecules. Different PAMPs activate different TLRs.
Toll-like receptor (TLR): a member of a class of ten innate immune receptors which recognize general chemical and structural motifs characteristic of pathogens.
Type I interferon (IFN): a class of cytokines which play a role in innate immune activation and orchestration of the adaptive T cell immune response.

Novel biomaterials allow researchers to target TLRs directly, but research on engineered TLR-targeting materials has been relatively non-systematic, often relying on trial-and-error approaches [8]. Development of immunomodulatory therapeutics is often conducted from the framework of DC targeting and activity [9]. In most ongoing work, the direct role of TLRs is not highlighted. With the recent advances in our understanding of this receptor class, it is important to leverage

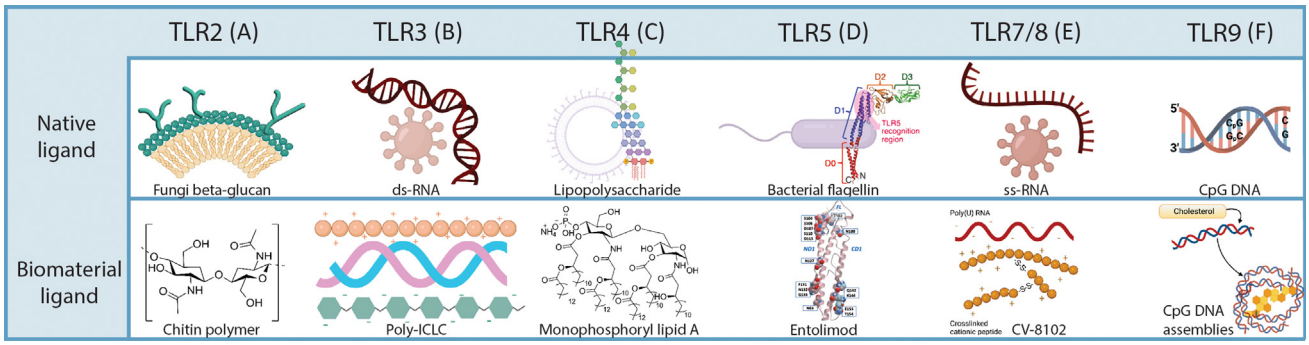


Figure 2. Examples of macromolecular Toll-like receptor (TLR) agents and assemblies. Natural and synthetic ligands, many of which mimic native or natural pathogen-associated molecular patterns (PAMPs), are used as tools to stimulate or suppress TLR-mediated immune activity. Assemblies based on these ligands or other synthetic polymers are used to adjuvant or deliver TLR agents; some assemblies are also shown to have inherent self-adjuvanting or modulatory properties. Macromolecular agonists are based on natural agonist ligands such as some nucleic acids, peptides, polysaccharides, and lipids. Upon recognition by a TLR (either at the cell surface or in the endosome), a series of signaling events occurs which alters cell cytokine production, further orchestrating innate and adaptive immunity. These can either be a proinflammatory immune state or an immunosuppressive state. (A) TLR2 is a receptor for bacterial lipids and yeast carbohydrates, but has been shown to display activity in response to chitin, pectin, and glucomannan. (B) TLR3 recognizes double-stranded viral RNA (dsRNA), and is stimulated by synthetic RNA-based adjuvant, Poly-ICLC, stabilized with polypeptide and polysaccharide molecules. (C) TLR4 is a receptor for lipopolysaccharide (LPS); this activity is leveraged clinically using synthetic analog monophosphoryl lipid A (MPLA). (D) TLR5's proinflammatory activity towards bacterial flagellin is recapitulated using entolimod, a synthetic recombinant protein. (E) TLR7/8 recognizes viral single-stranded RNA (ssRNA); notably, the CureVac CV-8102 complex uses synthetic poly(U) complexed to cationic peptide molecules. (F) CpG DNA, the native ligand for TLR9, can be fabricated into intricate assemblies which can enhance delivery or modulate TLR9 activity. (D) Reproduced under CC BY 4.0 (http://creativecommons.org/licenses/by/4.0/) [47,90]. (F) Assembly adapted, with permission, from [78]. Figure created with BioRender.

this knowledge towards immunoengineering to further enable the design and translation of safe yet efficacious therapies [10].

In this review, advances in biomaterials that interface with TLRs are discussed to provide an overview of recent developments in the field. This review is organized by ligand class, which generally corresponds with a particular TLR's activity. That is, biomaterials with similar structural features often have mechanistically similar ligand–receptor interactions. For each class, recently developed technologies for modulating TLR activity to tailor the immune response for a range of applications are discussed.

Synthetic polycations

Synthetic polycations include both oligomers and polymers in which the repeating unit possesses a protonatable nitrogen. At physiological pH, these amines are protonated to varying degrees depending on pKa and other factors.

Poly(ethylene imine)

Poly(ethylene imine) (PEI) has been used for a range of immunological applications; its cationic charge is often leveraged for immunotherapeutic scavenging of PAMPs to treat inflammatory disease (Figure 3B).

Sepsis, a systemic inflammatory response syndrome, is mechanistically driven by TLR activation via cell-free DNAs (cfDNAs). Hence, targeting cfDNA represents a potential strategy to treat sepsis. A 2020 study from Dawulieti *et al*. used nucleic acid-binding nanoparticles (NABNs) made from PEI-functionalized mesoporous silica as cfDNA scavengers [11]. Nanoparticles functionalized with PEI at varied charge density demonstrated stronger inhibition of inflammation at higher charge densities. This research highlighted the role of cfDNA in stimulating TLR9 and showed that

Key table

Table 1. Polymeric and macromolecular biomaterials interface with the TLR class of innate immune receptors with several modes of action

Macromolecule class	Identity	Action	Target	Refs
Synthetic polymer	Polycation	Scavenger	TLR9	[11,12,17]
			TLR4	[14]
			TLR 2, 3, 4, 9	[15]
		Antagonist conjugate	TLR2/6	[13]
		Agonist conjugate	TLR7/8	[16]
	Noncationic polymer	Antagonist	TLR4	[23]
		Agonist conjugate	TLR9	[24]
			TLR7/8	[25]
			TLR3, 4, 7/8, 9	[26]
			TLR2/6, 4	[27]
Lipid	LPS derivative	Agonist	TLR4	[28,30]
		Agonist conjugate	TLR4, 7/8	[32]
			TLR4	[33]
	Lipid assembly	Agonist delivery vehicle	TLR4	[29,31]
Protein and peptide	Lipopeptide	Agonist	TLR2/6	[37]
		Agonist conjugate	TLR2	[35,39,40]
		Agonist delivery vehicle	TLR4	[36]
			TLR7/8	[42,43]
		Agonist coupling	TLR2	[38]
	Peptide nanofibers	Agonist assembly	TLR2	[41]
	Glycopeptide	Antagonist	TLR4	[44]
	Antimicrobial peptide	Modulator	TLR9	[45]
	Flagellin mimetic	Agonist	TLR5	[46,47]
	Sericin	Antagonist	TLR4	[48]
	Poly(γ-glutamic acid)	Agonist	TLR4	[49–51]
Polysaccharide	Glycosaminoglycan	Antagonist	TLR4	[53]
			TLR2/4	[55]
		Agonist conjugate	TLR2	[54]
	Marine polysaccharide	Agonist	TLR2/4	[56]
			TLR4	[57,58]
		Scavenger	TLR2/4	[59,60,62]
	Chitin	Agonist	TLR2	[64]
		Modulator	TLR2	[65]
	Chitosan	Antagonist	TLR4	[63]
	Plant-derived polysaccharide	Agonist	TLR2	[66]
		Scavenger	TLR2	[67]
Nucleic acid	Oligodeoxynucleotide	Agonist	TLR9	[69–74]
		Antagonist	TLR4	[75]

Table 1. (continued)

Macromolecule class	Identity	Action	Target	Refs
	Nucleic acid nanostructure	Agonist	TLR9	[76–78]
		Antagonist	TLR2	[79]
		Modulator	TLR9	[80–82]
	RNA	Agonist	TLR3	[83]
		Agonist	TLR7/8	[84,85]

PEI NABNs neutralized cfDNA, prevented TLR activation, and mediated the inflammatory response. Further research utilizing this construct extends PEI's capacity as a scavenger to inflammatory bowel disease, attenuating TLR9 and demonstrating its potential as a therapy for several cfDNA-mediated diseases [12].

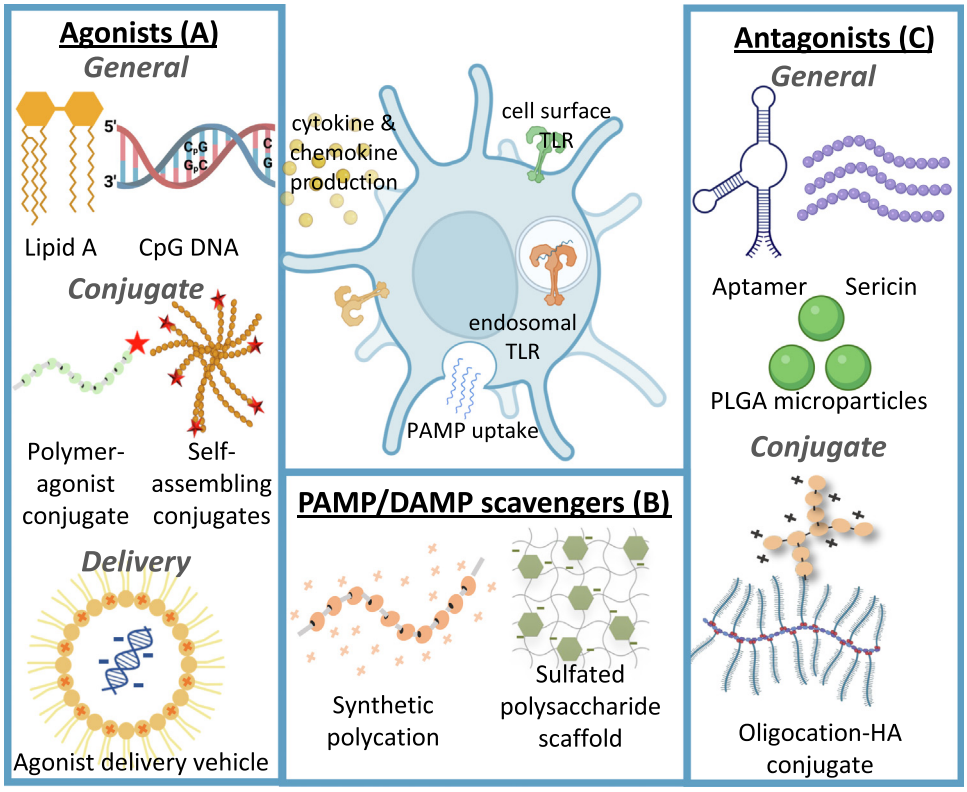


Figure 3. Toll-like receptor (TLR) agents can be classified as agonists, scavengers, and antagonists. This figure illustrates example macromolecules for the classes discussed in this review; a dendritic cell (DC) is shown. (A) Traditional TLR agonists include CpG DNA, lipid A, flagellin, lipopolysaccharide (LPS), and palmitoylated lipopeptides. Small-molecule agonists can be incorporated into macromolecular assemblies via conjugation to a polymer or lipopeptide [24,25,39,40]. Delivery vehicles can display inherent agonist activity, or can be used to deliver traditional TLR agonists [36]. (B) Materials designed to scavenge pathogen-associated molecular patterns (PAMPs) or damage-associated molecular patterns (DAMPs) typically consist of synthetic polycations which scavenge negatively charged materials; however, these can also be fabricated from sulfated polysaccharides which alleviate reactive oxygen species at the site of inflammation [62]. The scavenging of PAMPs/DAMPs indirectly attenuates TLR signaling. (C) Antagonist ligands of TLRs include siRNAs designed to antagonize TLRs [75], sericin protein [48], and certain glycopeptides [44] and glycosaminoglycans [53,55]. Abbreviations: HA, hyaluronic acid; PLGA, poly(lactic-co-glycolic acid). Figure created with BioRender.

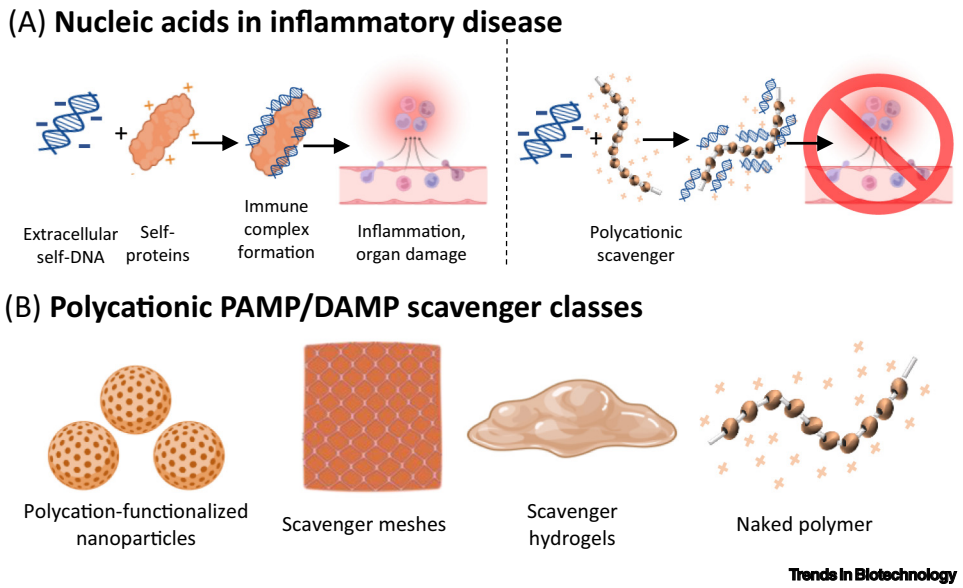


Figure 4. Polycationic agents are used to fabricate complex pathogen-associated molecular pattern (PAMP)/damage-associated molecular pattern (DAMP) scavengers. Cationic materials are capable of scavenging negatively charged PAMPs such as nucleic acids. (A) Extracellular nucleic acids are a hallmark of inflammatory disease and present upon aberrant cell death. These ligands interact electrostatically with cationic self-proteins, forming immune complexes which can accumulate in the organs or at sites of inflammation. When polycationic materials are used to scavenge PAMPs and DAMPs, a return to normal Toll-like receptor (TLR)-regulated immune activity is observed. (B) Polycationic materials can be used alone [17], or incorporated into scavenging constructs such as nanoparticles [11,12], meshes [15], or hydrogels [14]. The applications of polycationic scavengers are versatile, including the treatment of sepsis, inflammatory disease, traumatic injury, and organ dysfunction. Figure created with BioRender.

Oligoethyleneimines (OEIs) – shortened, branched PEI-derivatives – have been developed to supplement treatment for ulcerative colitis. A library of branched hybrid structures inhibited **nuclear factor κB (NF-κB)** signaling via several TLR pathways in a murine model of acute ulcerative colitis, most likely due to reactive oxygen species scavenging (Figure 3C) [13]. This effect was highly dependent on molecular architecture; TLR attenuation is optimized by striking a balance between low-**molecular-weight (MW)** OEI, which displays minimal positive charge, and high-MW OEI, which is highly cytotoxic.

Poly(amidoamine)

Poly(amidoamine) (PAMAM) dendrimers contain branched subunits of amide and amine functional groups. Due to their cationic charge, PAMAM dendrimers have shown utility in neutralizing negatively charged damage-associated molecular patterns (DAMPS), which trigger the immune system following trauma. The globular architecture of PAMAM, enabled by polymer chemistry techniques, increases charge display while minimizing toxicity, like branched OEIs. Shen *et al*. developed an immunoregulatory scaffold delivering PAMAM dendrimers for the treatment of spinal cord injury [14]. DAMPs released from injured tissue activate macrophages by binding to TLRs, upregulating proinflammatory cytokines which cause secondary damage. The cationic charge of the polymer scavenged and neutralized DAMPs and led to suppressed TLR4 expression. The modular scaffold also released regulatory cytokine interleukin 10 (IL-10), influencing macrophage polarization towards an anti-inflammatory phenotype, and showing promise in decreasing the secondary damage caused by spinal cord injury. Similar work uses a PAMAM-based

microfiber mesh for neutralizing DAMPs released following traumatic injury [15]. This strategy limits systemic exposure to PAMAM while promoting TLR ligand inhibition. PAMAM has also been used as a vaccine delivery platform, by fabricating star nanoparticles with a PAMAM dendrimer core and *N*-(2-hydroxypropyl) methacrylamide (HPMA) polymer arms extending outward. TLR7/8 agonist adjuvants and peptide immunogens were covalently attached to the HPMA arms, leading to improved immunogenicity [16].

Hexadimethrine bromide

Hexadimethrine bromide, commercially known as Polybrene, is a cationic polymer used to increase cellular transduction efficiency, and as an agonist to reverse heparin-induced anticoagulation. Researchers have demonstrated the novel use of Polybrene as a nucleic acid scavenger for applications in treating trauma-induced inflammation [17,18]. Tissue damage resulting from multiple organ dysfunction syndrome (MODS) releases DAMPs, including mitochondrial DNA (mtDNA), exacerbating organ damage. In 2018, Aswani *et al*. leveraged hexadimethrine bromide as a scavenging polymer to neutralize mtDNA and reduce organ damage in MODS [17]. They found that the scavenging activity correlated with rescued severe multiple organ injury in animal models, and attenuated NF-κB phosphorylation linked to TLR9. The potential of hexadimethrine bromide as a scavenger for cfDNAs is a promising solution to inflammation and injury following trauma.

Nucleic acids such as extracellular DNA trigger inflammatory disease (Figure 4A). In most cases, polycations act as PAMP/DAMP scavengers for anionic ligands of TLRs (DNA/RNA, ligands for TLR3, TLR7/8, and TLR9). However, this electrostatic interaction of polycations is not specific to the TLR ligands, as they can interact with any number of negatively charged biomolecules *in vitro* and *in vivo.* Furthermore, polycations are often cytotoxic due to the proton sponge effect, which may limit their use [19–21]. Some of this can be mitigated by engineering the physical presentation of the polycations (Figure 4B) into different forms, including nanoparticles, meshes, hydrogels, or free polymers. For example, both nanoparticles and naked polymers may be sufficiently small to be phagocytosed (and thus induce the proton sponge effect), but meshes and hydrogels may be sufficiently large to escape ingestion while still sequestering extracellular DAMPs.

Synthetic polymers

Polymers described in this section exhibit no charge on the repeating unit of the chain. These polymers typically act as carriers for, or conjugates of, TLR agonists or antagonists, but a direct TLR response is also possible.

Poly(lactic-co-glycolic acid)

Poly(lactic-co-glycolic acid) (PLGA)-based platforms can be tailored to modulate immune outcomes, dependent upon identity (i.e., lactide:glycolide ratio, MW) and architecture. While some reports investigate PLGA in the context of immune activation [22], PLGA-based microparticles have also been investigated for immunosuppression (Figure 3C) [23]. Allen *et al.* report that after administration, PLGA microparticles are phagocytosed and degraded, resulting in increased anti-inflammatory cytokine production. DCs undergo an immunosuppressive phenotype change via the NF-κB pathway after exposure to PLGA particles, which is associated with TLR activity. This activity was modulated by PLGA MW: smaller sizes led to greater immunosuppression. While this PLGA/NF-κB relationship is useful for immunoengineering, PLGA's status as a prominent biomaterial makes this differential behavior a potential consequence in nonimmune applications. Interestingly, the breakdown products of PLGA are negatively charged at physiological pH due to the carboxylic acid produced. Lower-MW PLGA breaks down to individual lactic acid and glycolic acid faster than higher-MW PLGA, implying that the production of these carboxylate groups may be implicated in its activity.

Poly(ethylene glycol)

The development of poly(ethylene glycol) (PEG)-based conjugates, or PEGylation, has been advantageous in the activation of TLR-expressing cells. A study by Wu *et al*. showed improved immunostimulatory activities through conjugation of PEG to CpG motifs (CpG–PEG). The CpG–PEG conjugate increased TLR9 activity in the tumor microenvironment of mice by increasing the circulation time [24]. PEG conjugated to the highly toxic TLR7/8 agonist resiquimod has also been shown to extend circulation time and reduce toxicity, while retaining the drug's immune activity *in vivo* (Figure 3A) [25]. In this case, the polymer is simply enhancing the pharmacokinetics of the TLR agonist.

Poly(styrene)

Poly(styrene) (PS) is not a known TLR agent, but its inert behavior makes it a useful material for the mechanistic analysis of TLRs. Electrospun PS microfibers, conjugated with other synthetic polymers (including cationic PEI and PAMAM-G3), have been used to study monocytic TLR tolerance in patients with coronavirus disease 2019 (COVID-19) [26]. TLR tolerance occurs when cells such as monocytes are repeatedly exposed to DAMPs and PAMPs, which leads to persistent and aberrant downstream TLR signaling [21]. This electrospun construct scavenged nucleic acid PAMPs, reducing the activity of TLR3, 4, 7/8, and 9, and elucidating how PAMP-induced TLR tolerance in COVID-19 patient blood samples affects patient outcome. PS enables the conjugation of polycations which are critical to the function of the mesh. PS has also been used in the mechanistic analysis of TLR response to agonist density and number [27]. The Esser-Kahn group developed an extensive library of PS microparticles conjugated to TLR agents to investigate different particle sizes and surface parameters, minimizing background immune activation through PS use.

Lipid-based macromolecules and assemblies

Lipids are large hydrophobic molecules that are typically smaller than most polymers and proteins, yet can assemble into larger structures or conjugates. Certain lipid ligands and assemblies have been engineered to interface with TLR4 as vaccine adjuvants.

LPS-derived macromolecules

LPS is a bacterial toxin comprising a hydrophobic lipid core conjugated to a polysaccharide chain. The LPS derivative monophosphoryl lipid A (MPLA) and its clinical analogs are used as anionic TLR4 agonists (Figures 2C and 3A). These synthetic derivatives have been utilized in recent reports investigating TLR activity in protein antigen vaccine applications [28–31]. Vaccines adjuvanted with these TLR4-activating materials elicit an enhanced cellular or humoral response to codelivered antigens. MPLA analogs have also been conjugated to small-molecule TLR agonists and synthetic peptides to improve vaccination efficacy through enhanced TLR signaling in seasonal influenza and cancer vaccination [32,33].

Lipid nanoassemblies

Lipid-based nanoassemblies solubilize, stabilize, and distribute active pharmaceutical ingredients. Formulations of commercialized and FDA-approved lipidic TLR4 adjuvants include liposome-based AS01, AS04, and RC529. Novel ionizable lipids are designed by anchoring hydrocarbon-rich moieties to pH-sensitive polymers [34]. Ionizable polymer-lipids fabricated into liposomes were conjugated to an additional targeting TLR2 agonist, h11c LP, and utilized as a platform for antitumor immunotherapy [35]. Additionally, lipid nanoassemblies can exhibit inherent TLR activity, and have been used as minimalist self-adjuvanting delivery agents. For example, a cationic lipid nanoparticle was developed for the delivery of mRNA for antitumor immunity (Figure 3A) [36]. The unloaded formulation was shown to stimulate the TLR4 signaling

pathway. This efficient platform combines adjuvant and delivery activities, potentially improving cargo capacity and DC targeting.

Protein- and peptide-based polymer assemblies

Proteins from nonmammalian sources can be antigenic, and in some cases recognized by TLRs directly, and synthetic peptides and proteins can be designed to directly interact with TLRs, as described in the following examples.

Peptides (LPs, glycopeptides, and antimicrobial peptides)

The ability of LPs to act as antigens or adjuvants makes them promising candidates for vaccine development. Synthetic LP adjuvants are widely used in studies to interact with TLR1, 2, and 6; these are typically fabricated via conjugation of at least one hydrocarbon chain to the N-terminal or cysteine residue of the peptide.

A synthetic analog of a *Mycoplasma fermentans* LP, macrophage-activating lipopeptide-2s (MALP-2s), was shown to bind and interact with TLR2/6. Takeda *et al*. demonstrated the potential for this compound as an adjuvant, along with tumor-associated antigens, as a synergistic antitumor immunotherapy [37]. Other synthetic LPs enable the development of self-adjuvanting vaccines via covalent bioconjugation or electrostatic coupling, which have been studied for the treatment of *Mycobacterium* infection [38–40]. Recent work has also seen the fabrication of conjugates of LPs and protein antigens that can self-assemble into nanofiber constructs used for intranasal vaccines, retaining the LP's ability to activate TLR2 and offering an adjuvanted delivery platform for disease treatment [41]. Conversely, the Seder group developed a personalized cancer vaccine platform using charge-modified peptide-TLR7/8 agonist conjugates that self-assemble into nanoparticles [42,43]. These platforms utilize charge modification to program self-assembly with reduced variability compared with lipid-modified peptides.

By contrast, certain peptides display immunoinhibitory or modulatory behavior. Zein-derived glycopeptides, found in corn kernels, have been enzymatically modified with carbohydrates to induce hepatoprotective effects in alcohol-induced liver injury mouse models. These peptides were found to protect against elevated TLR4 mRNA expression levels [44]. Another notable example is the antimicrobial peptide LL37, which self-assembles with nucleic acids to form complexes which modulate TLR9 activity, dependent on biophysical structure [45].

Flagellin mimetics

Bacterial flagellin is recognized by TLR5 during infection. Entolimod, a synthetic protein derived from *Salmonella* flagellin, is a TLR5 agonist that also has immune-stabilizing properties (Figure 2D) [46]. Most recent entolimod research focuses on its utilization as a radiation countermeasure; however, a small number of studies have developed second-generation derivatives of entolimod in cancer immunotherapies [47].

Sericin-based polymers

Sericin is a protein primarily derived from the silkworm (*Bombyx mori*). While sericin has been widely used for tissue engineering scaffolds, it is also known for displaying anti-inflammatory properties (Figure 3C). Sericin hydrogels have been utilized for their anti-inflammatory effects in the treatment of myocardial infarctions. A study by Song *et al.* analyzed the downstream effects of sericin on reducing proinflammatory tumor necrosis factor α (TNF-α) expression [48]. These studies showed the ability of injectable sericin hydrogel to suppress TLR4, playing an anti-inflammatory role by interfering with TLR-mediated signaling pathways.

Poly-γ-glutamic acid

Poly-γ-glutamic acid (γ-PGA), manufactured by bacterial fermentation with *Bacillus* sp., has been shown to stimulate TLR4. Lee *et al.* demonstrated the use of γ-PGA to increase **type I interferon (IFN)** production mediated via TLR4 and its accessory factors. The group reported viral load clearance in mouse models after oral administration of γ-PGA [49]. γ-PGA can be further fabricated into nanoparticles with chitosan, which work synergistically to induce immune responses via TLR4 [50].

Comparative studies show that γ-PGA derived from different *Bacillus* species have different effects on TLR recognition. Jelacic *et al.* showed that γ-PGA derived from *B. subtilis* and *Bacillus licheniformis* elicit higher TLR2 recognition. Similarly, *Bacillus anthracis*-derived γ-PGA was shown to produce the highest recognition in TLR4-expressing HEK cells. These results suggest that the isomeric composition of γ-PGA can affect TLR recognition [51]. Differences such as these could be leveraged for tailored immune outcomes.

Designing peptides and proteins to mimic known ligands for TLRs from the microbial world could result in therapies to amplify innate immune responses, which may be desirable in applications such as vaccine design or as adjuvants in immunotherapies.

Polysaccharides

Several classes of natural polysaccharides are known to modulate the immune system. In most cases, the polysaccharides interact with TLR4, for dual-directional regulation resulting in either immunosuppression or immunoactivation. The natural ligand for TLR4 is LPS (Figure 2) which is an outer membrane component of Gram-negative bacteria. Bacterial glycans have a significantly greater diversity in monosaccharide composition, saccharide modification, and linkage patterns than mammalian glycans, and thus many prokaryotic glycans (and fungal eukaryotic glycans) can be distinguished by the human immune system on the basis of sequence and/or structural features [52]. Polysaccharides can also interact with TLR2, which detects β-glycans on fungi. Given the wide diversity of glycans in nature, it is no surprise that natural and synthetic oligo- and polysaccharides may be recognized by receptors on innate immune cells. These biopolymers may also exhibit desirable mechanical properties, biocompatibility, and/or low-toxicity, making them ideal for many medical applications.

Glycosaminoglycans

Endogenous polysaccharides have been shown to regulate both innate and adaptive immunity. **Glycosaminoglycans (GAGs)** often interact with TLR4 in their native forms; however, they have not been extensively engineered for immunomodulation. One notable exception is heparin-derivatized nanoparticles that suppress TLR4 [53]. The biophysical parameters of the self-assembled particle library and their consequences on TLR activity were studied using spectroscopic techniques and molecular dynamics simulations. TLR suppression correlated with long aliphatic chains and the presence of sulfate groups, which enhanced binding strength.

Another endogenous GAG, hyaluronic acid (HA), can have both pro- and anti-inflammatory properties, depending on its MW. High-MW HA is generally known for its anti-inflammatory properties, and low-MW HA is preferentially recognized by TLR2 or TLR4, triggering proinflammatory cytokine production. Thus, HA conjugates have recently been explored as an alternative to PEGylated vaccine adjuvants, activating the TLR2 pathway [54]. Conversely, Huang *et al*. found that low-MW HA may also be used to elicit TLR-mediated anti-inflammatory effects when modified with β-azide or β-methyl groups [55]. This discrepancy is not well understood,

but may be dependent on cell type. As our understanding of HA/TLR activity evolves, this relationship can be better leveraged towards engineering applications.

Marine polysaccharides

Polysaccharides derived from algae have immunostimulatory properties that enhance phagocytosis and cytokine expression. Galactans, for example, trigger inflammation via the TLR2/4 signaling pathway [56]. Similarly, alginate is a prominent marine-derived biomaterial that is well suited for tissue engineering and drug delivery. Studies have linked alginate to the upregulation of TLR4 [57]. This relationship has been harnessed via a drug-loadable calcium alginate hydrogel for oral tissue repair [58]. Following treatment with the unloaded hydrogel, human periodontal ligament cell expression of proinflammatory cytokines, TLR4, and osteogenesis-related genes increased. In both examples, increased expression of TLR4 coincided with increased inflammation, implying the involvement of the TLR4/NF-κB pathway. This widely utilized biomaterial provides inherent immunostimulatory benefits for regenerative medicine.

Alternatively, algae-derived fucoidans have been established as inhibitors of TLR2/4, shown through structural studies of their dose-dependent PAMP scavenging (Figure 3B) [59,60]. Their immune properties are retained even when functionalized with photopolymerizable groups for tissue engineering applications [61]. The Zhu group fabricated fucoidan-loaded nanofibrous scaffolds for intervertebral disc reconstruction; the entrapped fucoidan suppressed TLR-mediated expression of proinflammatory genes [62]. Marine plants are a rich source of natural polysaccharides, marked by several classes which can be leveraged for different outcomes in immunoengineering.

Chitin and chitosan

Chitin and its deacylated analog chitosan both serve as potential TLR agents. Chitin triggers a TLR2-dependent immune response (Figure 2A), while the immune response to chitosan primarily occurs via TLR4 [63]. Recently, chitin was used in the development of a vaccine against *Coccidioides*, or San Joaquin Valley fever [64]. A recombinant antigen encapsulated in hybrid β-glucan/chitin particles significantly decreased colony-forming units (CFUs) of *Coccidoides* and stimulated a proinflammatory response via TLR2, potentially aided by the dual-receptor targeting of β-glucan and chitin. Like HA, the TLR activity of chitin is directly related to oligomer size, providing a target for engineering activity [65].

Plant-derived polysaccharides

Glucomannan is a plant-derived polysaccharide exhibiting structural similarities to PAMPs, which typically present glucan or mannan groups that are recognized by TLRs. Glucomannan was functionalized with acetyl groups, mimicking the aliphatic groups present in natural immunostimulatory microbial signals [66]. Increasing the degree of acetylation of the glucomannan derivative led to an M1 phenotypic change in macrophages, and modulated TLR2 signaling to increase proinflammatory cytokines. The acetyl-modified glucomannan resulted in proinflammatory signaling which reduced tumor growth *in vivo*.

The fruit-derived polysaccharide pectin has also been utilized as a TLR2 modulator. The modification of pectin with different degrees of methyl esterification enables its use as a DAMP scavenger [67]. The polymer was embedded into a microcapsule for pancreatic islet implants in diabetes treatment, and showed reduced TLR2 activation and long-term implant survival in an animal model. While the data indicate that TLR2 attenuation is related to prevention of DAMP-induced immune activation, the mechanism is unclear, as the pectin most effective at preventing immune activation had the lowest degree of esterification as well as the lowest MW.

Nucleic-acid-based polymers and assemblies

Synthetic and natural nucleic acids (NAs) are biopolymer ligands for TLR3, 7, 8, and 9. Natural cfDNAs have implications in autoimmune and degenerative diseases [68]. However, this relationship can also be harnessed for modulating TLR-mediated immune activity.

Oligodeoxynucleotides and aptamers

Many synthetic **oligodeoxynucleotides (ODNs)** have been classified as TLR agonists due to the unmethylated CpG motifs displayed on their surface (a PAMP recognized by TLR9) (Figure 3A). As of 2022, only one ODN has been FDA-approved and commercialized. This TLR9 immunostimulatory sequence (ISS), CpG 1018, is currently used as an adjuvant for hepatitis B surface antigen, and has recently undergone clinical investigation for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) vaccines globally [69,70]. Other ODNs are used in cancer research: for example, ODN SD-101 which is enrolled in various Phase 1 and 2 clinical trials [71]. Preclinically, CpG ODNs have been investigated in combination with antitumor mRNA vaccines, chemotherapeutics, and ultrasound ablation [72–74]. Aptameric DNAs, which interface with TLRs with high specificity, can be designed as either agonists or antagonists. Recently, an aptameric TLR4 antagonist was developed to treat brain ischemia reperfusion in an animal model (Figure 3C) [75].

NA nanostructures

The immunological activity of NA assemblies with programmable architecture is partly dependent on their higher-order structure, enabling a direct interface with immune cells, enhanced stability and delivery of DNA, and spatial control of TLR activity. Chemical modification and conjugation have enabled the manufacture and delivery of CpG DNA nanostructures for TLR9 stimulation [76]. Nanoparticles and spherical structures containing TLR9 adjuvant DNA have shown promise in immune stimulation for cancer immunotherapy [77]. Modifications using hydrophobic lipid moieties confer increased stability of the construct (Figure 2F) [78]. Conversely, immunoinhibition is achieved using siRNA nanostructures, which have been shown to downregulate TLR2 in sepsis applications [79]. While nucleic acids typically do not interface with TLR2, siRNAs block the expression and activation of TLR2 via gene silencing. The different modes of action of these structures is based on NA identity, but additional modifications can create a more tailored immune response.

Recent efforts to uncover the structure–property relationships of NA assemblies and resulting immunological activity will improve clinical translation. Tunable TLR binding can be achieved through modulation of structural stability, and, as recently shown by Comberlato *et al*., inter-ligand distances [80]. Engineered DNA-based biomaterial assemblies provide a bottom-up approach to studying the structure–property relationships of both naturally occurring inflammatory NA complexes and engineered constructs with TLRs [81]. This is of particular importance in the case of immune polyelectrolyte multilayers (iPEMs), which typically involve the delivery of charged nucleic acids [82]. The potent TLR activity of NAs, combined with a toolbox of modification methods and evidence of overcoming translational hurdles, make them a logical material for interrogating innate immunity.

RNA-derived TLR agents

Poly-ICLC, a synthetic complex of carboxymethylcellulose, polyinosinic-polycytidylic acid, and poly-L-lysine double-stranded RNA (dsRNA) mimic, has shown promising Phase 1/2 results as a TLR3 adjuvant in a combination therapy for patients with melanoma (Figure 2B) [83]. Synthetic single-stranded RNA (ssRNA) derivatives display TLR7/8 agonistic behavior which has been harnessed as a vaccine adjuvant. Recently, CV8102 has shown promising results from Phase 1/2 clinical trials for hepatocellular carcinoma (Figure 2E) [84]. While most NA assemblies utilize

CpG DNA to target TLR9, an RNA/DNA hydrogel developed by the Nishikawa group targets TLR7 and 8 through the controlled release of RNA from the gel. This construct confers increased stability to its RNA payload [85].

Concluding remarks and future perspectives

TLR-targeting macromolecules often mimic native TLR ligands, offering a pattern recognition opportunity that is specific and tunable. These macromolecules and their TLR activity are outlined in Table 1. Engineered platforms based on higher-order macromolecular assemblies and combinatorial approaches enhance immunoengineering and vaccine efficacy [86]. Tools founded in bioconjugate chemistry, synthetic biology, nanoengineering, and polymer chemistry provide a basis for the modification of these ligands for immunoengineering. These tools enable engineers to uniquely coalesce the natural and synthetic to design novel immunoengineering strategies. Vishwakarma *et al*. previously identified a trend in the increased development of 'new generations of biomaterials' – which serve as hybrids of native and synthetic structures – for immunoactive applications [87]. This is seen in the mimicking and modulation of the native TLR ligand as a template in therapeutic design, which is considered the best-developed strategy for TLR modulation, FDA approval, and clinical translation [4]. Despite the advancements, the dearth of knowledge of the role TLRs play in disease biology has contributed to the failure of TLR drugs in clinical trial, highlighting a continued need for mechanistic analysis to improve engineering approaches (see Outstanding questions).

Macromolecules can be engineered to modulate, enhance, or enable TLR activity via several general approaches – including conjugation of polymers to TLR agents, coassembly of macromolecules with TLR agents, development of drug delivery agents such as liposomes for TLR agents, and larger-scale constructs such as polymer gels that can interact with and sequester TLR agents – and computational modeling can be used to guide their design (Figure 5). Macromolecules and macromolecular assemblies provide several unique advantages over small-molecule ligands for interrogating TLRs. First, macromolecules and macromolecular assemblies have the ability to exhibit multiple copies of a ligand at a defined spacing (Figure 6A,B), allowing the engagement and clustering of multiple TLRs simultaneously [88]. Second, the activity of a macromolecular ligand in TLR signaling can be tailored through relatively simple alteration to the macromolecule, such as increasing or decreasing the degree of modification or MW (Figure 6C,D). Third, macromolecules and macromolecular assemblies can exhibit controlled 3D shape and size on a scale relevant to the size and geometry of TLRs [89]. Fourth, macromolecules and macromolecular assemblies can exhibit favorable pharmacodynamics compared with small-molecule therapeutics, as they can protect cargo from protein adsorption or degradation, can exhibit increased circulation lifetimes, and can be functionalized to target specific populations of cells or altered physiology (i.e., reduced pH seen in inflammation).

While the utilization of TLR agents has increased, the current trial-and-error approach of TLR agent synthesis or discovery may limit the clinical translation of therapeutics. Elucidating the relationship between biophysical properties and immune activity will advance drug design [3]. Bottom-up approaches utilizing computational tools, molecular dynamics simulations, high-throughput screening techniques, biophysical experiments, and crystallography will expand the toolbox of TLR-targeting therapeutics. Applying these technologies for structure–activity studies with libraries of macromolecular mimetics of native ligands will enable further structural understandings and guide predictive drug design. With this improved mechanistic understanding, TLR ligands can be rationally engineered for more nuanced approaches to immunomodulation, broadening the scope of treatable diseases and improving current cancer and vaccination strategies.

Outstanding questions

How can computational technologies enable more rational design or modification of macromolecules for targeting TLRs?

Is TLR activation an important consideration in the choice of biomaterials for nonimmune applications? Can we tailor current strategies to avoid off-target immune activation using material properties such as particle size or porosity?

How can the spatial display of PAMPs in nanoassemblies be leveraged to enhance TLR activity? Can biophysical modeling be used to more rationally design drug delivery vehicles and self-assembled structures of interest?

How can novel engineering tools such as click chemistry be leveraged by bioengineers to generate tailorable platforms of TLR agents? Could large libraries of TLR-interfacing macromolecules be utilized to generate a more mechanistic understanding of TLR activity?

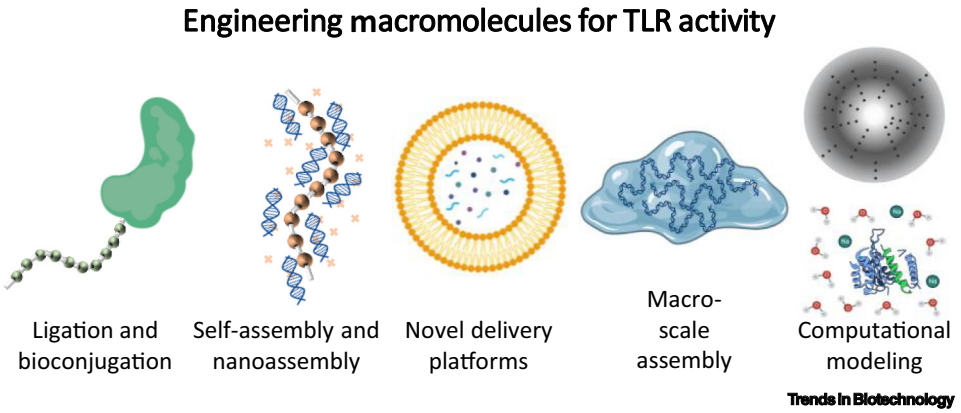


Figure 5. A representation of engineering techniques which enable optimization or interrogation of Toll-like receptor (TLR) activation. This review provides an overview of techniques which modify existing TLR ligands, as well as the discovery of TLR agents that are not prominently used as biomaterials. To increase the activity of TLR ligands or coalesce the activities of two bioactive molecules, covalent bioconjugation can be used to design novel compounds which copresent to immune cells. Self-assembly of charged polyelectrolytes optimizes their delivery, but can also be leveraged for modulation of TLR signaling. These nanoscale delivery platforms can display inherent TLR activity, dependent on the macromolecules that are used to fabricate the vehicle. Larger constructs, such as scaffolds, can be used to scavenge pathogen-associated molecular patterns (PAMPs) or damage-associated molecular patterns (DAMPs), or can be used for tissue engineering to promote certain immune outcomes. Libraries of TLR biomaterial assemblies can be used to probe the activities of TLRs, particularly in the context of biophysics. When these biophysical experiments are combined with computational tools such as molecular dynamics simulation, a more foundational understanding of TLRs can be achieved which can guide more predictive design of future ligands and constructs. Figure created with BioRender.

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Declaration of interests

No interests are declared by the authors.

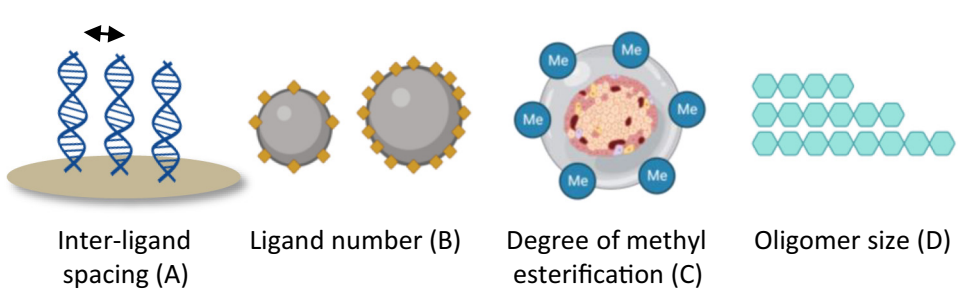


Figure 6. Macromolecules provide tunable, modular platforms for tailored immunoengineering. Strategies engineering agonist and antagonist ligands can be used to tune Toll-like receptor (TLR) response. TLR activity can be optimized via (A) ligand spacing [80], (B) ligand number [27], (C) hydrophobicity [67], (D) size [65], and charge [82]. Further investigation and optimization of these parameters will inform TLR agent design. Figure created with BioRender.

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